Replication of 'People can use the placement of objects to infer communicative goals' by Lopez-Brau & Jara-Ettinger (2023, Cognition)

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Introduction

The main goal of this project is to replicate the results from Experiment 2 of the Lopez-Brau & Ettinger (2023) paper. A secondary goal of this project is to partially replicate the results from Experiment 3a of the same paper. The paper broadly revolves around the inferences humans make about the social meaning of objects, specifically investigating whether adults are able to infer the communicative meaning of objects from their placement. Experiment 2

finds that adults believe that objects that impose a low cost on deciders are more likely to be communicative relative to objects that do not impose a cost. Experiment 3 furthers these findings: adults infer that a low-cost object in front of a door serves as a signal communicated by the person who placed the object that others should avoid that door. I anticipate leading projects on similar questions to investigate what inferences children make about the social meaning of objects. Replicating the methods and analysis will therefore be beneficial, serving as background findings motivating my studies and as a first step to brainstorming potential project designs and analysis plans for my own studies.

There will be two parts of the replication project. The first part will include stimuli that consists of eight sets of images. Each set will have a pair of doors side-by-side: one "low-cost door" (object placed in front of the door) and one "no-cost door" (object placed beside the door). The objects will be a plant, a chair, a pile of books, a pile of cinderblocks, some tape, some meter sticks, a hat, and a fishbowl tied to a tack on a door frame. Participants will be asked to imagine leaving an office and encountering this pair of doors. They will be asked a series of multiple-choice questions: an attention-check ("What objects are in front of the doors?"), test trial ("Which door do you think someone was trying to tell you something?"), a manipulation check ("Which door requires more work to walk through?") and lastly, an inclusion question ("Do you think you would be able to walk through this door if you wanted to?"). These methods will match the procedure from Experiment 2 of the paper. The second part will include stimuli that consists of 8 more images of pairs of doors, with each pair consisting of an empty door and a door with an object placed directly in the middle of one of the doorways. Half of these door pairs will be open and the other half will be closed. The objects will be the same as listed above and participants will again be asked to imagine leaving an office and encountering this pair of doors. Participants will then answer: an attention-check ("What is the only difference between the two exits?"), test trial ("What do you think someone was trying to tell you about the door with the object?"; the options provided will be (a) "You should walk through the door with the object" and (b) "You should not walk through the door with the object"), manipulation-check (same as Exp 2), an inclusion questions (same as Exp 2). These methods are a shortened version of the procedure from Experiment 3a of the paper (only including the "low-cost" condition) to account for time constraints of running the replication on SONA.

I anticipate the main challenges to be setting up the paradigm, specifically programming it using JavaScript which is unfamiliar to me. I imagine that it might be further challenging to code the study design such that failure on the attention checks reroutes participants to the start of the story and does not allow them to continue with the experiment until they pass the check (as per the published study design). I also anticipate collecting data from SONA to be challenging as the process is new to me. I hope to learn more about coding a jsPsych experiment and collecting data from an adult sample. Gaining skills in carrying out a jsPsych experiment and collecting data on SONA may both be helpful in the case I need to run or pilot studies on adult samples. Given some familiarity I have with running behavioral studies along with transparent materials and code provided by the authors, I am optimistic that this project will be feasible over the course of this quarter. More importantly, I hope to gain familiarity

with how to replicate findings from a published project: from finding relevant materials and files from the original paper, to recreating the methods for data collection, all the way to rerunning the original analyses on my dataset and writing them up.

Link to my repository: https://github.com/anlaha/lopez-brau2023; Original Paper hosted on repository at: https://github.com/anlaha/lopez-brau2023/tree/main/original_paper

Methods

Power Analysis

Original effect size, power analysis for samples to achieve 80%, 90%, 95% power to detect that effect size. Considerations of feasibility for selecting planned sample size.

Planned Sample

Planned sample size and/or termination rule, sampling frame, known demographics if any, preselection rules if any.

Materials

All materials - can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Procedure

Can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Analysis Plan

Can also quote directly, though it is less often spelled out effectively for an analysis strategy section. The key is to report an analysis strategy that is as close to the original - data cleaning rules, data exclusion rules, covariates, etc. - as possible.

Clarify key analysis of interest here You can also pre-specify additional analyses you plan to do.

Differences from Original Study

Explicitly describe known differences in sample, setting, procedure, and analysis plan from original study. The goal, of course, is to minimize those differences, but differences will inevitably occur. Also, note whether such differences are anticipated to make a difference based on claims in the original article or subsequent published research on the conditions for obtaining the effect.

Methods Addendum (Post Data Collection)

You can comment this section out prior to final report with data collection.

Actual Sample

Sample size, demographics, data exclusions based on rules spelled out in analysis plan

Differences from pre-data collection methods plan

Any differences from what was described as the original plan, or "none".

Results

Data preparation

Data preparation following the analysis plan.

Confirmatory analysis

The analyses as specified in the analysis plan.

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Replication Attempt

Open the discussion section with a paragraph summarizing the primary result from the confirmatory analysis and the assessment of whether it replicated, partially replicated, or failed to replicate the original result.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis, (b) assessment of the meaning of the replication (or not) - e.g., for a failure to replicate, are the differences between original and present study ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the replication attempt. None of these need to be long.